



Usage of the Concept of Organizational Capability in the Japanese Quality Management Context

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Abstract: This paper elucidates how the concept of “organizational capability”—which tends to fall into a tautology and for which it is difficult to establish a unified scale in academic empirical research—is used in the Japanese practical context. Therefore, we clarified the “meaning attributed to the concept of organizational capability” within Japanese Total Quality Management (TQM), which actively employed the term “organizational capability” to contribute to enhancing the international competitiveness of Japanese companies. As a result, it became evident that Japanese-style TQM standardized the “procedures” through which each company defines, implements, and evaluates organizational capability in accordance with its own context and not the “contents” of organizational capability itself. It also became clear that the organizational capability realized differs across companies. This can be viewed as a practical response by Japanese-style TQM to

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the research problem that assessing organizational capability retroactively based on outcomes tends to fall into a tautology. Therefore, in organizational capability research, rather than applying a uniform measure, it is necessary to carefully ascertain the organizational capability of each company through dialogue with practitioners while referencing standardized procedures.

Keywords: organizational capability, TQM, quality management, manufacturing

Introduction

Organizational capability is an organizational factor that has attracted attention as a source of competitive advantage that cannot be reduced to mere resources (Collis, 1994; Fujimoto, 1997; Grant, 1996; Helfat & Peteraf, 2003; Henderson & Cockburn, 1994; Kogut & Zander, 1992; Leonard-Barton, 1992). Grant (2013), one of the standard texts on strategy, states that a company's organizational capability is formed by linking together the various resources it possesses, and cites Toyota Motor Corporation's lean production system, which integrates various capabilities ranging from the manufacturing of parts and subassemblies to supply chain management, quality management, and continuous improvement, as an excellent example.

Fujimoto (1997), who empirically elucidated the organizational capability of Toyota Motor Corporation, views organizational capability as the ability to generate a competitive and stable flow by utilizing given management resources. Furthermore, Fujimoto, Amano, and Shintaku (2007) define manufacturing organizational capability as a pattern of resource combination and utilization (a bundle of routines) specific to individual companies for executing the

process of creating, transferring, and transmitting design information toward the customer—which serves as the foundation for a company’s survival and prosperity—more accurately, efficiently, rapidly, and continuously than competitors. Organizational capability is an important concept in management studies, but its measurement is not easy because it can fall into a tautology, making it difficult to apply a unified scale (Grant & Verona, 2015).

This paper aims to clarify how this academically important concept is utilized in the Japanese business community and what effects it is considered to have. Specifically, we identify how the concept of organizational capability is used in a practical context based on Japanese quality management, Total Quality Management (TQM), and the Deming Prize, which actively use the term “organizational capability” to promote improvements in corporate competitiveness, as well as descriptions in JIS standards regarding quality management. Furthermore, we examine the initiatives of Toyota Motor Kyushu as an example of defining and measuring its own organizational capability. As a result, it becomes clear that in Japanese-style TQM, it was not the “contents of organizational capability” that were standardized, but rather the “procedures by which each company defines, implements, and evaluates its own organizational capability.” This can be positioned as a practical response by Japanese-style TQM to the problem in organizational capability research that capturing organizational capability by working backward from results tends to fall into a tautology. The research implication derived from this practical response is that the competitive advantage of a given company is acquired and maintained due to the uniqueness and distinctiveness of the routines constituting its organizational capability and the ways in which they are combined. Therefore, it is necessary to design a measurement method with the understanding that while the contents of these routines can be relativized in comparison with those of other

companies, they cannot be standardized.

Usage of the Concept of Organizational Capability in Japanese Quality Management

1. Quality control and TQM in Japan

After World War II, organizations such as the Union of Japanese Scientists and Engineers (JUSE) played a central role in promoting the introduction of statistical quality control (SQC) from the United States and Japanese Total Quality Control (TQC) to Japanese companies, which significantly contributed to improving the competitiveness of Japanese manufacturing companies. QC circles (frontline small-group activities) were deployed on a nationwide scale and, from the 1960s onward, developed into all-employee participation-based TQC (Company-Wide Quality Control, CWQC), involving development, purchasing, marketing, and other functions. TQC was a quality control activity unique to Japanese companies, characterized by a broad scope of vertical involvement from top management to the frontline and cross-functional horizontal involvement. In the 1970s and 1980s, the manufacturing quality (conformance quality) of Japanese products reached the world's top level, and the QC circles and TQC that supported this, as well as the "Deming Prize" established in 1951, were highly evaluated internationally (Fujimoto, 2001; Takahashi, 2016).

In the 1980s, the United States sought to restore its competitiveness by learning from the TQC of Japanese companies and, in 1987, established the Malcolm Baldrige National Quality Award (MB Award), which benchmarked the Deming Prize. Within a few years of its establishment, the award significantly changed the awareness and behavior of U.S. managers, functioning as a guideline for quality improvement that made them recognize the importance of quality and customer satisfaction (Garvin, 1991). The Malcolm

Baldrige National Quality Award (MB Award) evaluates the quality of underlying management activities (management quality) rather than product quality, and differs from the Deming Prize in that it includes not only process management but also the formulation of management visions and strategies (TQM Committee, 1998). In Japan as well, the Japan Productivity Center established the Japanese version of the MB Award, the “Japan Quality Award,” in 1996 (Takahashi, 2016).

In Western countries, TQM became widespread rather than TQC, which was strongly characterized by product quality control. In response to this, Japan’s TQC was also renamed TQM. JUSE (“TQM Declaration,” 1997) changed the conventional “Total Quality Control” to “Total Quality Management” in April 1996, intending to develop it as a new form of TQC while leveraging the strengths of Japanese-style TQC.

Subsequently, the international standardization of quality management systems, represented by the ISO 9000 series, advanced. The series is a standard concerning the “quality system” that purchasers require from suppliers—namely, the supplier’s formalization and documentation of its own quality management processes as internal rules (procedures and instructions)—and customers trust a company equipped with this as “a company with quality management capability” (Fujimoto, 2001).

In the next section, we will examine how the concept of “organizational capability” is used in these contexts.

2. Organizational capability in quality management system standards and the Deming Prize

(1) The concept of organizational capability in Japan’s early TQM

We review the situation in the late 1990s, when the renaming to TQM took place, based on the TQM Committee (1998). TQM is a management science and management technology that aims to

improve the quality of management processes (business processes, value chain) and the quality of the management resources (people, equipment, organization, information, knowledge, technology, etc.) associated with them. In this context, companies and organizations aim for a “respected presence” and a “relationship capable of sharing inspiration with stakeholders,” seeking to improve their “admired competitiveness,” namely their “technological capability,” “responsiveness,” and “vitality.” This “praised competitiveness” was considered to be “organizational capability,” which corresponds to a company’s core competence. Improving this capability was precisely the direct objective of TQM, and it was assumed that profits would naturally follow.

When promoting TQM, self-assessment is recommended, and for “technological capability,” “responsiveness,” and “vitality,” respectively, assessment items such as possession of high technological capability and a technology strategy, rapid decision-making, and entrepreneurship at the management level are presented (TQM Committee, 1998).

The TQM Committee (1998) positioned the introduction of the new concept of organizational capability to improve the quality of the management system as a point of differentiation not found in the Malcolm Baldrige National Quality Award (MB Award), and assumed that through these activities, organizational capability would be accumulated and become a company’s core competence. However, the pathways through which the improvement of organizational capability leads to the realization of a “respected existence” and “good relationships,” as well as where the management system should be improved to enhance organizational capability and how TQM can contribute to this, were identified as issues that were not explicitly addressed.

As described above, in the late 1990s, the renaming of TQC as TQM and the restructuring of concepts were advanced, and the

improvement of organizational capability was established as the objective of TQM. However, how to specifically define and measure its contents and link them to the management system was not sufficiently operationalized and remained an unresolved issue. However, in JIS Q 9005, which is discussed in the next section, the focus shifted from “identifying the contents of organizational capability” to “the procedures by which each company conceives and implements its organizational capability profile.”

(2) The concept of organizational capability in JIS Q 9005

We examine the concept of organizational capability in the Japanese Industrial Standard “JIS Q 9005,” which was established as a Japanese response to the ISO 9000 series. JIS Q 9005/9006 were established in December 2005; JIS Q 9005 specifies a quality management system model for continuing sustainable growth in response to environmental changes, while JIS Q 9006 provides guidelines on methods to support self-assessment of its maturity, along with improvement and innovation (Iizuka, 2006).

In JIS Q 9005, based on business strategy and policy, organizations are required to clarify their organizational capability profile, which is the overall picture of the capabilities they should possess, and to build and operate a quality management system suitable for their own organization. That is, customer value is defined based on the value of products and services, and the organizational capability profile is established as a set of key capabilities that are decisive for competitive advantage. After identifying strengths and weaknesses through a gap analysis with the existing situation, an analysis is conducted to determine which quality management system elements should incorporate these capabilities, aiming for improvement and innovation through strengthening these elements. In the methodology set out in JIS Q 9006, self-assessments are conducted by establishing self-assessment criteria for the

elements to be emphasized, and the results are used as inputs for strategic management review to drive innovation.

Therefore, JIS Q 9005 defines the organizational capability profile as “the overall picture of organizational capability necessary for competitive advantage that should be possessed in the future in order to secure competitive advantage and continuously provide products and services.”

Following the establishment of the first edition in 2005, JIS Q 9005 was revised in 2014 and 2023 (the latest edition is JIS Q 9005:2023). In JIS Q 9005:2023, emphasis is placed on the sequence of identifying the organizational capability necessary for providing value to customers and society, and planning and designing a quality management system as a means to implement it (Kaneko, Iizuka, & Yamada, 2023). Providing value superior to that of competitors requires recognizing the company’s own characteristics that serve as the source of its capabilities and utilizing and strengthening them as organizational capabilities. The storyline of which characteristics to utilize in which organizational capabilities and what kind of value to provide constitutes the business scenario. The standard defines the organizational capability profile (organizational profile) as the overall picture of the capabilities that an organization should possess and that are necessary for providing value to customers and society for sustained success. Here, capability refers to the state of being able to perform something. To ensure that these capabilities are reliably demonstrated, it is necessary to identify the key quality management system elements, define the attributes and states these elements must possess as well as the interrelationships among them, and clarify the elements that embody the learning capability required to respond to change.

Additionally, the standard cites (1) “specific technologies, knowledge,” (2) “planning and design, effectiveness review, and innovation of the quality management system,” (3) “product and

service realization process,” and (4) “operation and management of management resources” as elements of the quality management system.

Comparing JIS Q 9005 with ISO 9001 (JIS Q 9001 and 9004), the concept of organizational capability is present in JIS but absent from ISO, and methods for learning capability, improvement, and self-assessment were also not adopted in ISO despite the efforts of the Japanese side (Kaneko et al., 2023). The Japanese approach, which emphasizes improvement and learning, was not accepted in international standards. However, it is a strength of Japanese companies that overseas companies cannot easily imitate, and it can become a source of sustainable competitive advantage.

Furthermore, Item 16.9 of the Japanese Society for Quality Control standard “Quality Management Terms (JSQC-Std 00-001:2023)” defines organizational capability as “the ability of an organization to perform specific activities.” Specific activities include functional activities such as planning, design and development, procurement, and manufacturing, as well as cross-functional management activities such as quality, cost, and delivery. Organizational capability is demonstrated through activities and, in a broad sense, includes organizational culture (values and behavioral patterns shared among members).¹ A clear definition of the elements of organizational capability and their levels is called an organizational capability profile.

As described above, what was standardized in JIS Q 9005 was not the specific contents of organizational capability, but the procedures by which each company conceptualizes an organizational capability profile in line with its own value proposition and business scenario, implements it into the QMS, and evaluates and improves it. In this

¹ In 16.10 of the same standard, organizational culture is defined as the values and behavioral patterns shared among the members of the organization.

respect, it can be said that the concept of organizational capability has been repositioned from an abstract objective concept in early TQM to a practical management target defined and operated by each company itself.

(3) The concept of organizational capability in the Deming Prize

We examine how the concept of organizational capability is used in the Deming Prize, which contributed to the dissemination of postwar quality control activities, based on its examination criteria. According to the Deming Prize Committee of JUSE, the Deming Prize provides guidelines for utilizing TQM to realize management objectives and strategies, and the process of applying for and receiving the prize has been utilized as an opportunity to achieve a dramatic improvement in organizational capability. The examination does not seek conformity to a given model. Rather, it evaluates the results and processes of company-wide improvements and reforms that the organization itself has undertaken after recognizing its situation and setting issues and objectives, as well as their effectiveness over the long term. This fundamental principle of an “examination reflecting the situation of each organization” has remained unchanged since its establishment in 1951. Recent awards have expanded to the overseas bases of Japanese companies and to overseas companies.

The evaluation criteria for the Deming Prize have been revised several times, and the current criteria are the 2018 revised edition. Prior to that, the criteria revised in 2002 for the purpose of clarifying the “examination viewpoints” were used through fiscal year 2017.

The Deming Prize is said to be awarded to applicant organizations that satisfy the following three conditions (“the three conditions A–C for receiving the Deming Prize”) through TQM:

- A) In accordance with the management philosophy, industry,

business format, scale, and business environment, management objectives and strategies that are proactively customer-oriented and furthermore take into account the social responsibility of the organization are formulated based on clear management intent. In addition, top management demonstrates leadership in their formulation.

- B) TQM is appropriately utilized and implemented toward the realization of A's management objectives and strategies.
- C) As a result of B, effects have been achieved regarding A's management objectives and strategies, and the organizational capability necessary for future development has been acquired.

The criteria through fiscal year 2017 were “Basic Items,” “Distinctive Activities,” and “Role of Top Management and its Fulfillment,” but their correspondence with the three conditions ABC was ambiguous. Therefore, in the 2018 revision, these were reorganized into a format that completely corresponds to the three conditions ABC, and “supply chain,” “accumulation and utilization of knowledge,” and “social responsibility” were added.

Items related to organizational capability were positioned, until fiscal 2017, as “Organizational Strength” (maintenance and enhancement of core technology, speed, and vitality), an item under “Role of Top Management and Its Execution,” and from fiscal 2018 onward, as “Acquisition of Distinctive Activities and Organizational Capability,” an item under “Effects of TQM.” In the latter, the consistency among management objectives and strategies, TQM activities, their results, and the acquisition of organizational capability necessary for future development is examined, and the award report summaries also mention tangible and intangible effects almost without exception. In other words, the activities and initiatives that lead to high performance constitute “organizational capability” in TQM and the Deming Prize, and although its contents may vary by

company, it has become possible to evaluate each company's initiatives based on a (somewhat standardized) process of definition, implementation, and outcome verification.

In discussions on quality management in Japan, the activity elements of TQM correspond to the basic elements of organizational capability. These consist of (1) new product development management and process assurance for planning and designing products and services based on customer needs, and producing and providing them exactly as designed; (2) policy management, small-group improvement activities, and quality control education for formulating, deploying, and implementing strategies for improvement initiatives within the organization; and (3) standardization and daily management for ensuring that each department reliably fulfills its roles regarding regular operations not covered by policy management (Nakajo & Yamada, 2006).

Hosotani and Iwasaki (2019) cite multifaceted intangible effects of TQM activities, in addition to the tangible effects of QCDSME (quality, cost, delivery, safety, morale, environment), such as the establishment of a robust corporate constitution, the company-wide deployment of top management policies, customer trust gain through the permeation of "quality first," the embedding of fact-based management and PDCA, human resource development through QC circles, and the elimination of sectionalism. As tangible effects, it is reported that 3 to 4 years after introducing TQM, sales and profits increase by 1.5–3 times, process defect rates and complaint amounts decrease to 1/10 to 1/100, and lead times decrease to 1/2 to 1/3; these findings are obtained through the improvement of organizational capability via TQM activities.

Hosotani and Iwasaki (2019) state that as a condition for receiving the Deming Prize, the organizational capability to realize management objectives and strategies is given the highest importance; they also find that organizing the three elements of

management strategy, organizational capability, and company activities with competitive advantage using a “T-shaped matrix” clarifies existing capabilities, capabilities to be newly acquired, and the direction for strengthening organizational capability.

As described above, in the Deming Prize, the specific contents of organizational capability are not uniformly defined. Rather, what is evaluated is whether each company can define the necessary organizational capability in light of its management objectives and

Table 1. Transition of the concept of organizational capability and quality-related awards in TQM

Period	Concepts of organizational capability and the organizational capability profile in quality control and TQM	Quality-related awards
1951	—	The Deming Prize is established by the Union of Japanese Scientists and Engineers (JUSE). The examination evaluates the results and processes of the improvements and reforms the organization has carried out company-wide—with issues and objectives set by the organization itself—and their future effectiveness (“examination that reflects each organization’s situation”).
1987	—	The United States establishes the Malcolm Baldrige National Quality Award (MB Award), benchmarked against the Deming Prize. It evaluates the quality of management activities (management quality), encompassing not only process control but also the formulation of management vision and strategy.
Late 1990s	Organizational capability = “admired competitiveness” (“technological capability,” “responsiveness,” and “vitality”). Its enhancement is the direct objective of TQM. Self-diagnostic items are also provided.	The Japan Productivity Center establishes the Japan Quality Award, a Japanese version of the MB Award (1996).
2002	Examination item related to organizational capability: positioned as “organizational strength” (the maintenance and reinforcement of core technologies, speed, and vitality) under “the role of top management and its fulfillment” (through FY2017).	The Deming Prize examination criteria are revised (to make the “examination viewpoints” easier to understand; used through FY2017).
2005	JIS Q 9005 introduces the concept of the “organizational capability profile,” defined as “the overall picture of the organizational capabilities necessary for competitive advantage that the organization should possess in the future in order to secure competitive advantage and to provide products and services on a continuing basis.” Each firm clarifies its profile based on its business strategy and policy, conducts a gap analysis against the current state, implements the important capabilities in QMS elements, and pursues improvement and innovation by reinforcing those elements.	—
2018	The item related to organizational capability moves to “distinctive activities and the acquisition of organizational capability” under “the effects of TQM” in the Deming Prize. Emphasis is placed on the consistency running from management objectives and strategies, through TQM activities and their effects, to the acquisition of the organizational capability necessary for future development.	The Deming Prize examination criteria are revised (current version). The former “basic items,” “distinctive activities,” and “the role of top management and its fulfillment” are reorganized to correspond to the three conditions A–C for the prize. “Supply chain,” “accumulation and utilization of knowledge,” and “social responsibility” are added.
2023	• JIS Q 9005:2023: defines the organizational profile as “the overall picture of the capabilities the organization should possess to provide value to customers and society for sustained success.” A capability = “a state in which something can be done”; emphasis is placed on first identifying the organizational capability required to provide value, and then planning and designing the QMS as the means of implementing it. • JSQC standard: defines organizational capability as “the ability of an organization to carry out specific activities” and the organizational capability profile as a clarification of the elements of organizational capability and their respective levels.	—

Source: Author

strategies, implement it into its TQM activities, and explain it by linking it to outcomes and future development potential. The concept of organizational capability (organizational capability profile) and the evolution of quality-related awards in Japanese quality management and TQM can be summarized as shown in Table 1. The next section examines Toyota Motor Kyushu as a case of defining and indexing company-specific organizational capability under such a framework.

3. The promotion of TQM and operationalization of organizational capabilities: The case of Toyota Motor Kyushu

What kind of organizational capability is acquired through the pursuit of TQM and the Deming Prize? We examine the initiatives of Toyota Motor Kyushu, which received the Deming Prize in 2016 and the Deming Grand Prize in 2019, based on Yoneoka and Nakamura (2020), who were involved in promoting these initiatives.

Toyota Motor Kyushu (hereafter, TMK) is a wholly owned manufacturing subsidiary of Toyota Motor Corporation established in 1991 that develops and produces Lexus vehicles under Lexus International Company. Through its pursuit of both awards, various TQM activities were conducted, and the improvement of organizational capability was positioned as one of the pillars of strategy execution.

At the beginning of the examination, the concept of organizational capability was ambiguous, and the capabilities possessed by each organization had not been organized. At the time, President Nihashi communicated the necessity of high frontline, technological, and human capabilities as driving forces for improving the competitiveness of the site and “retaining manufacturing in Kyushu,” and these were incorporated into TQM activities as the “driving force.” However, initially, this remained limited to an “individual development initiative” focused on the number of professional and multi-skilled personnel developed.

Subsequently, organizational capability improvement was positioned as a means of executing strategic themes and, while being related to the existing initiatives of each department, efforts to define organizational capability and develop indicators advanced from around 2017. A mechanism for improvement was established as the “8 Steps for Improving Organizational Capability.”²

TMK uniquely defined organizational capability and established its three components as the “frontline capability” possessed by the entire workplace, and the individual capabilities that foster it, namely “technical capability” (an aggregate of technical skills) and “human capability” (an aggregate of human skills) (Table 2).

“Technical capabilities” and “human capabilities” are aggregates of individual capabilities, and based on existing tools such as job capability evaluations and shop-specific capability requirement sheets, the capabilities to be added or strengthened and the number of personnel to be developed to possess them are determined. “Frontline capability” is assumed to improve not only through individual technical and human capabilities but also through improvements in “mechanisms, environments, and structures,” and its constituent elements are broadly classified into technical and skill aspects and interpersonal relations aspects (Table 2).

As measures to enhance these individual and workplace capabilities, TMK assumes two approaches: improving “environments, mechanisms, and structures” and “development

² These eight steps include (1) clarifying the elements of organizational capability (the relation between technological and human capabilities that fosters frontline capability), (2) developing indicators for each capability element of frontline capability, (3) setting targets for frontline capability, (4) developing indicators for technological capability and human capability, (5) setting targets for technological and human capabilities, (6) examining measures to improve organizational capability, (7) executing organizational capability improvement measures, and (8) evaluating organizational capability and confirming its effectiveness.

Table 2. Overview of the elements of organizational capability at Toyota Motor Kyushu

Category	Item	Description
Technical capability	Problem/issue-solving capability	The ability to solve problems and issues.
	Failure-prevention capability	Failure-prevention capability utilizing FMEA, FTA, QA, QCMS, risk assessment, quality engineering, etc.
	Strategy-building capability	The ability to formulate vision, strategy, and tactics.
	Planning and creativity (technical)	The planning and creative ability to develop products, new processes, and methods.
	Professional/multi-skilled personnel	Technical track: professional and multi-skilled personnel in design and production engineering. Skill track: advanced skills and multi-skilling.
Human capability	Leadership	Leadership of executives, line management, and project leaders.
	Motivation	Individual motivation and satisfaction.
Genba (frontline) capability	Improvement capability (team)	The ability to make improvements as a team.
	Core competence (technical)	The ability to design—from styling and product planning through detailed design—and to mass-production at a high level (performance, quality, weight, cost, etc.).
	Innovation capability	The ability to develop new products and groundbreaking new processes and methods, and to innovate and transform processes, etc.
	Information-gathering and analysis capability	The ability to gather and analyze information.
	Teamwork	Teamwork in the workplace.
	Coordination capability	Coordination across functions, departments, and stakeholders.
	Speed	Rapid decision-making and execution, change, and transformation.
	Climate of improvement and challenge	An atmosphere and culture of improvement and challenge.

Source: Prepared by the author based on Yoneoka and Nakamura (2020), Table 4.3.1.

through education, training, etc.” The former includes small-group activities, development environments, industry–academia–government collaboration, and ingenuity and patent systems, among others.

These capability elements were linked to the detailed strategic themes of the site. The formulation of indicators, considered the most difficult task, was carried out collaboratively by the corporate planning, human resources development, technology development, and production engineering departments, while examining the ideal future state of frontline capabilities with reference to factors such as changes in the business environment and government materials. Utilizing existing internal data (such as the level of QC circles and survey results on workplace management and employee satisfaction), the maturity of each capability element was defined and

Table 3. Overview of the measurement indicators of organizational capability at Toyota Motor Kyushu

Category	Element	Measurement method	Measurement indicator
Technical capability	Problem/issue-solving capability	Quantitative & qualitative	QC circle individual evaluation and job-competency evaluation (average value).
	Failure-prevention capability	Qualitative	Newly established. Satisfaction rate (the proportion of people at levels 4 and 5 among the 1–5 levels of individual capability evaluation).
	Strategy-building capability	Qualitative	Job-competency evaluation. Satisfaction rate (the proportion of people at levels 4 and 5 among the 1–5 levels of individual capability evaluation).
	Planning and creativity (technical)	Qualitative	Newly established. Satisfaction rate (the proportion of people at levels 4 and 5 among the 1–5 levels of individual capability evaluation).
	Professional/multi-skilled personnel	Quantitative	Newly established items and the capability requirement sheets used by each department. Satisfaction rate.
Human capability	Leadership	Quantitative & qualitative	Job-competency evaluation (average value).
	Motivation	Quantitative & qualitative	Workplace management questionnaire and employee attitude survey (average value).
Genba (Frontline) capability	Improvement capability (team)	Quantitative	Scoring of circle levels (e.g., "tackle" activities in the skill track and "scrum" activities in the office track).
	Core competence (technical)	Qualitative	Five-point maturity evaluation of the following items: quality commitment, concurrent engineering, resolution of the core technologies needed to satisfy customer needs, the capability to realize and mass-produce product plans and concepts, mastery of the design-development and production-engineering environment, and technology-development capability.
	Innovation capability	Qualitative	Five-point maturity evaluation of the following items: top-management leadership, innovation strategy, innovation process, and open innovation (collaboration with external parties).
	Information-gathering and analysis capability	Qualitative	Details not specified.
	Teamwork	Quantitative	Affirmative-response rate in the workplace management questionnaire.
	Coordination capability	Quantitative & qualitative	Details not specified.
	Speed	Quantitative & qualitative	Five-point maturity evaluation of the following items: speed of decision-making on strategy and measures; transmission, sharing, and permeation of information; timely resource input; organizational and personnel structures; execution and action; and process transformation.
	Climate of improvement and challenge	Quantitative	Affirmative-response rate in the workplace management questionnaire and participation rate in the creativity-and-ingenuity proposal system.

Source: Prepared by the author based on Yoneoka and Nakamura (2020), Tables 4.3.2, 4.3.3(a), and 4.3.3(b).

evaluated on a 5-point scale for the new qualitative indicators, and elements combining qualitative and quantitative measures were also established (Table 3).

Each capability element organized in this way was evaluated using the established indicators; target values for 2020, 2025, and 2030 were set for frontline capability, technical capability, and human capability; and improvement measures to close the gap with the current state were examined and implemented. However, according to Yoneoka and Nakamura (2020), this initiative was at the stage of being trialed in specific model workplaces, and company-wide deployment, systematization of the mechanism, and linkage with the human resources system remained unresolved issues. What the TMK

case shows is not that the categories of frontline capability, technical capability, and human capability are organizational capabilities common to all companies. Rather, what is important is that the company defined its organizational capability in light of its strategy and desired state and translated it into measurement metrics, target values, and improvement measures. This case demonstrates how the process of conceiving, implementing, and evaluating a company-specific organizational capability profile, envisioned in JIS Q 9005 and the Deming Prize, is materialized within a company.

Discussion and Conclusion

This article examined how the concept of organizational capability is used in practice, based on JIS standards, the assessment criteria of TQM and the Deming Prize, and cases of award-winning companies.

In Japanese quality management and TQM, while aligning with ISO 9001 and 9004, JIS standards were established as guidelines that reflect the realities of Japanese-style manufacturing and value creation, and organizational capability and organizational capability profile are positioned as core concepts. However, what JIS provides is limited to the definition of organizational capability, and its specific contents are to be conceived in accordance with each company's strategy and the value it seeks to create.

As an example of a Deming Prize-winning company that internally defined and measured organizational capability, we examined the initiatives of Toyota Motor Kyushu. Aiming for the Deming Prize and the Deming Grand Prize, the company defined and measured the organizational capability of its site in light of its strategy and desired state, and worked to improve and evolve it. Taking on the challenge as an opportunity to thoroughly consider what the company's organizational capability is led to an improvement in competitiveness. Conversely, without such strong motivation, the

identification and measurement of invisible organizational capability are unlikely to be actively pursued and are not easy for researchers, let alone practitioners.

Based on the above, the contribution of this paper is to clarify that what was standardized in Japanese-style TQM is not the contents of organizational capability itself, but the procedures by which each company defines, implements, and evaluates organizational capability in accordance with its own context. This can be positioned as a practical response by Japanese-style TQM to the research problem that assessing organizational capability retroactively from outcomes tends to fall into a tautology. Therefore, the organizational capability profile envisioned by practitioners can differ from company to company. While it is reasonable in itself to regard unique organizational strengths that generate unique customer value as organizational capability, when researchers conduct inter-firm comparisons or benchmarking, they must be mindful of the diversity of content that practitioners associate with this term.

Takahashi (2010) breaks down the concept of “organizational strength” into two elements: (1) the capability to organize people into a single cohesive entity (“our internal organizational strength”), and (2) the greater power exerted by uniting as an organization (“outward-facing organizational strength”). Generally, “organizational capability” refers to “outward-facing organizational strength,” but when management scholars and practitioners ask, “What is organizational capability?”, what is actually sought is an explanation of the state of “our internal organizational strength.” In light of this argument by Takahashi (2010), “organizational capability” in Japanese TQM refers to “our internal organizational strength” in contexts where tangible and intangible effects are evaluated, and the TQM activity elements for realizing this are positioned as initiatives to enhance “our internal organizational strength.” Therefore, when referring to enhancing organizational capability through TQM, both practitioners and

researchers must carefully consider which of the two is meant.

Regarding the challenges in empirical research on organizational capability, Grant and Verona (2015) identify three problems in quantitative research. The first is the issue of measurement scales; because it is difficult for respondents to accurately know the actual conditions of competitors in subjective evaluations, the use of objective “factual” information is recommended. The second is the problem of approximating unobservable organizational capability with observable variables, and using the number of new product launches or research and development investment as a proxy lacks conceptual validity. The third is the problem of treating organizational capability as a black box and using indirect indicators to explain performance; while assuming “increased experience → capability enhancement → performance improvement,” it actually analyzes “increased experience → performance improvement,” failing to elucidate the capability itself.

However, in qualitative research, identifying organizational capability is key, and it is essential to appropriately establish research protocols and clarify the identification criteria. In doing so, it is necessary to define in advance and comprehend, as the micro-foundations of organizational capability, the outcomes (performance) of specific tasks and functions, the cognition of the actors (who demonstrated the capability and for what reasons), and observable behaviors (decision-making, activities, and processes). Grant and Verona (2015) indicate that overcoming these challenges requires persistent empirical research that references excellent studies and pays attention to research design and measurement methods.

Based on the above, in empirical analyses of organizational capability targeting Japanese companies, rather than applying a uniform scale, it is necessary to design research, with reference to standardized procedures, on the premise of the diversity of practitioners’ conceptions. If researchers do not carefully define the

meaning of the term “organizational capability,” dialogue with practitioners will fail to align, and results with high validity and relevance cannot be obtained. Therefore, it is indispensable to accurately understand each company’s organizational capability through careful dialogue between practitioners and researchers and to devise research methods that enable this.

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